

# Physics-Informed Multi-PMU Event Detection and Localization: On-Site Hackathon Report (SGSMA 2026, IEEE 39-Bus)

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SGSMA 2026 Synchrophasor Anomaly Detection Competition — On-Site Round

**Abstract**—We apply our pre-trained, CPU-only physics-informed ExtraTrees pipeline to the on-site dataset `RAW_onsite`. The method aligns the provided PMU streams, builds baseline-relative electrical and data-quality features over fixed 30 s windows, and routes each window through hierarchical detection, event classification, and typed localization. This report summarizes the dataset, the feature set (including features added on site), and the detection, classification, and localization results.

## I. METHOD (UNCHANGED FROM THE SUBMITTED PIPELINE)

The deployed model is bus-agnostic: PMUs are discovered from file names (`Bus<N>_*.csv`), so the same artifact runs on a new bus placement without retraining. Each PMU stream is sorted by `TIMESTAMP` and split into fixed 30 s windows. Per window we compute baseline-relative magnitude, angle, frequency/ROCOF, derivative, robust  $z$ -score, sustained-anomaly, data-quality, power-proxy, and graph-temporal features ( $\approx 45k$  columns). A hierarchical ExtraTrees stack then predicts: (i) abnormal vs. normal, (ii) the event class  $\{0-8\}$  via a physical classifier with physics gates for bad-data (7) and missing (5/6), and (iii) the origin through typed `BUS/LINE/PMU` localizers. Output is the official 4-column CSV (`TIMESTAMP`, `Bus`, `Predicted_Event`, `Predicted_Location`).

## II. ON-SITE DATASET `RAW_onsite`

The provided package contains `[TODO]` observed buses (`[TODO]` PMU CSVs),  $\approx [TODO]$  rows each spanning  $\approx [TODO]$  minutes at 30 fps, yielding `[TODO]` analysis windows. Event classes observed in the labels: `[TODO]`. Timestamps are aligned by row index across PMUs (an outer merge on `TIMESTAMP` is avoided because of sub-ns drift between files).

## III. FEATURES

Table I lists the feature families used. For this dataset the working feature count is `[TODO]`. **Added on site:** `[TODO]` (describe any feature added for this dataset).

## IV. RESULTS ON `RAW_onsite`

Table II reports detection, classification, and localization on the on-site data, next to the submitted-model baselines for

TABLE I  
FEATURE FAMILIES FED TO THE EXTRATREES STACK.

| Family                         | What it captures                           |
|--------------------------------|--------------------------------------------|
| Segment statistics             | onset / transient / settling magnitude     |
| Temporal deltas & derivatives  | abruptness, ramps, post-event shift        |
| Robust $z$ -score              | scale-normalized abnormality               |
| Sustained anomaly              | duration above threshold (event vs. spike) |
| Data quality                   | NaNs, gaps, missing-run length             |
| Single-signal / bad-data score | isolated channel faults                    |
| Power proxy ( $P, Q$ )         | rough active/reactive movement             |
| Graph-temporal                 | PMU-pair spread weighted by elec. distance |

TABLE II  
METRICS: SUBMITTED BASELINES VS. ON-SITE DATASET.

| Metric                        | <code>RAW0001</code> | <code>SIM</code> | <code>RAW_onsite</code> |
|-------------------------------|----------------------|------------------|-------------------------|
| Detection accuracy            | 1.0000               | 0.9693           | <code>[TODO]</code>     |
| Abnormal recall               | 1.0000               | 0.9860           | <code>[TODO]</code>     |
| Event classification accuracy | 1.0000               | 0.9673           | <code>[TODO]</code>     |
| Event classification macro-F1 | 1.0000               | 0.9598           | <code>[TODO]</code>     |
| Localization Top-1 accuracy   | 0.8333               | 0.8524           | <code>[TODO]</code>     |

reference. Confusion matrices and accuracy curves are shown in Figs. 1–2.

## V. EFFICIENCY

The runtime is CPU-only with **0 trainable neural parameters** and a model bundle under **1 MB**. Inference on `RAW_onsite` took `[TODO]` minutes on a laptop CPU (macOS). No GPU, no feature scaling; NaNs are handled by a median imputer inside each model.

## VI. DISCUSSION

The pipeline transfers without retraining because features are baseline-relative and bus-agnostic. Expected sensitivities on unseen data: (i) localization Top-1 depends on how many candidate buses are observable; (ii) Event 8 (unknown) is used when no compact signature matches, which trades recall on rare classes for precision; (iii) bad-data (7) and missing (5/6) are decided by physics gates before the physical classifier, so communication artifacts do not contaminate physical-event labels.

Detection / event confusion matrix on RAW\_onsite  
`\includegraphics here`

Fig. 1. Confusion matrix on the on-site dataset. Replace the box with `fig_cm.pdf` once generated.

Per-class precision / recall / F1 and accuracy bars  
`\includegraphics here`

Fig. 2. Per-class metrics and overall accuracy on RAW\_onsite.